
Python Operators Homework Questions

Part 1: Arithmetic Operators

1. What are the five basic arithmetic operators in Python? List them and provide a brief description of what each does.
2. Given `a = 10` and `b = 3`, what will be the result of the following operations?
 - `a + b`
 - `a - b`
 - `a * b`
 - `a / b`
 - `a % b`
 - `a // b`
 - `a ** b`
3. Explain the difference between the division operator (`/`) and the floor division operator (`//`). Provide an example for each.
4. What is the purpose of the modulo operator (`%`)? Give a practical example where it might be used.
5. Calculate the result of the following expression, showing your steps: `5 + 2 * 3 - 8 / 4`.
6. How does Python handle operator precedence in arithmetic expressions? Use an example to illustrate.
7. Write a Python program that takes two numbers as input from the user and prints their sum, difference, product, quotient, remainder, and power (first number raised to the second number).

Part 2: Relational (Comparison) Operators

8. List the six relational operators in Python. What is the data type of the result when you use a relational operator?
9. Given `x = 25` and `y = 15`, evaluate the following expressions and state whether the result is `True` or `False`:
 - `x > y`
 - `x < y`
 - `x == y`
 - `x != y`
 - `x >= y`
 - `x <= y`
10. Explain the difference between `==` and `=`. Why is it important not to confuse them?
11. Can you compare strings using relational operators? If yes, how does Python determine the result (e.g., `'apple' > 'banana'`)?
12. Write a Python program that asks the user to enter their age. If the age is 18 or greater, print "You are an adult." Otherwise, print "You are a minor."
13. Create a Python program that checks if a given number is positive, negative, or zero using relational operators and `if-elif-else` statements.

Part 3: Assignment Operators

14. What is the primary purpose of assignment operators in Python?
15. What is the shorthand assignment operator for addition? Provide an example of how to use it.
16. Given `value = 100`, write Python code using assignment operators to:
 - Increase `value` by 20.
 - Decrease `value` by 15.
 - Multiply `value` by 2.
 - Divide `value` by 5 (using floor division).
 - Calculate the remainder of `value` divided by 7 and assign it back to `value`.
 - Raise `value` to the power of 3.
17. Explain the benefit of using compound assignment operators (e.g., `+=`, `-=`) over their longer forms (e.g., `a = a + b`).
18. Can you use assignment operators with strings? If so, provide an example using `+=`.
19. Write a Python program that initializes a variable `total` to 0. Then, using assignment operators, add 5, then 10, then subtract 3, and finally multiply by 2. Print the `total` after each operation.
20. True or False: `x += 5` is the same as `x = x + 5`. Justify your answer.

Challenge Questions:

21. Combine your knowledge of all three types of operators to write a Python program that calculates the area of a circle. The program should:
 - Prompt the user to enter the radius.
 - Use the appropriate arithmetic operator for squaring the radius.
 - Use a relational operator to check if the entered radius is positive. If not, print an error message.
 - Use an assignment operator to store the calculated area.
 - Print the area formatted to two decimal places. (Hint: Use `math.pi` or approximate pi as 3.14159)
22. Consider the following Python code:

```
p = 7
```

```
q = 3
r = p % q
s = p // q
t = r ** s
print(t >= (p - q))
```

Without running the code, predict the output and explain your reasoning step-by-step.
